

SONAM BHARTI

+91 7903810871 | s.bharti14021999@gmail.com | sonam-portfolio-three.vercel.app
linkedin.com/in/sonam-bharti-213127178 | github.com/sonambharti | medium.com/@sonam.iitjammu2023

Skills

Programming Languages: Python, Java, C++, JavaScript, HTML, CSS
Databases & Storage: DynamoDB, MySQL, MongoDB, Redis, AWS S3
Core CS Fundamentals: Data Structures & Algorithms, OOP, System Design, Operating Systems, Networking
Frameworks & Libraries: Node.js, React.js, FastAPI, Flask, Numpy, Pandas, TensorFlow, PyTorch
Vector Databases & Retrieval: AWS Knowledge Base, OpenSearch Serverless, Weaviate, ChromaDB, Pinecone, FAISS
Agentic AI: Prompt Engineering, RAG, Multi-Agent Workflows, Google ADK, Function Tools
GenAI Tools: Langchain, HuggingFace, OpenAI, GROQ API, Cursor, Antigravity
Cloud & DevOps: AWS (EC2, ECR, Amplify), GCP, Docker, Kubernetes, Microservices, REST APIs, Grafana, Jira
Developer Tools & Practices: Linux, Git, Postman, CI/CD, Agile Methodologies, Code Reviews, Debugging

Experience

Full Stack Software Developer | Vibrium AI , Hyderabad Aug 2025 – Present

- **Configured and deployed LLM-based Voice Bots** for 4 enterprise clients using **prompt engineering**, persona design, and **engineering contracts** across 5+ conversational AI use cases.
- **Architected scalable RAG pipelines** using AWS Knowledge Bases, Titan Embeddings, and OpenSearch Vector DB with retrieval latency below 40ms for enterprise AI assistants.
- Enhanced **website features, reusable libraries, and WhatsApp Business API integrations** to enable automated scalable omnichannel communication and significantly improved business outreach and operational efficiency.
- **Led** the end-to-end design and deployment of a **WebSocket-based** system for **real-time audio streaming**, successfully onboarding 2 clients for seamless media communication.
- Engineered **real-time AI** speech pipelines integrating **LLM, ASR**, and **TTS** over **WebSockets** achieving sub-1s latency, with full-duplex streaming enabling barge-in interruptions for natural conversational experiences.
- Built a hybrid **AI assistant sdk, packaged as an npm module** using **webRTC, LLM orchestration**, and **tool calling** enabling UI automation, and accessibility-driven interactions through natural voice and text commands.
- **Collaborated directly with enterprise clients** to gather requirements, define technical solutions, and deliver scalable AI systems aligned with business objectives.
- **Remediated VAPT findings and enforced secure SDLC governance** through secure coding standards, audit-compliant review processes, and coordination of enterprise compliance audits with clients and vendors.

Tools: *Python, Next.js, Typescript, Websocket, Livekit, JWT, Git, Jira, Prompt Engineering, LLM, Langchain, OpenAI, Weaviate, Pinecone, Dynamo DB, PostgreSQL, RDS, AWS (EC2, S3, Bedrock, Knowledge Bases, OpenSearch Service, Titan, Amplify, Lambda, Cloudwatch).*

Full Stack Developer (SDE-I) | Saarthi.ai , Bengaluru Aug 2023 – Jul 2025

- **Led development of conversational AI and NLU systems** for 15+ BFSI clients; solely developed and owned **fraud detection** bots using scalable NLP pipelines, and **SOLID** - compliant **production-grade code**.
- **Built RAG-based Generative AI applications** using LangChain, vector databases (like **Weaviate, Chroma**), and prompt engineering, reducing bot deployment lifecycle from 5 days to 1 day.
- Developed **scalable REST APIs**, using **CRUD** functionality , and built microservices and for enterprise integrations, enabling secure and robust ETL workflows between client infrastructure and internal systems, ensuring seamless data integration and transformation.
- **Automated data preprocessing** using Python scripts, reducing processing time by 40%, and **trained CUX team** in dataset creation, which improved NLU and NER accuracy by 20% for BFSI clients.
- Implemented solutions to identify **new entities through NER**, trained, **fine-tuned and deployed NER/NLU models** across 3 verticals, and managed production workloads on AWS cloud infrastructure using Docker containers.
- **Optimized high-traffic voice and WhatsApp bots in Agile methodologies** using asynchronous APIs and distributed architectures, reducing latency by 30% and infrastructure costs by 40%.
- **Performed production debugging and observability analysis** using Grafana and MongoDB performance tuning, improving uptime and reducing support tickets by 50%.
- **Containerized and orchestrated applications** using Docker, Kubernetes, and CI/CD pipelines for reliable large - scale deployments, while **reviewing code** to ensure best practices in accuracy, and efficiency.
- **Collaborated with cross-functional teams** to streamline development workflows and, **authored API and systems documentation**, and contributed to product architecture, improving maintainability, onboarding, and delivery efficiency.

Tools: *Python, FastAPI, Flask, PyTorch, Javascript, Node.js, RESTful API, AWS, Docker, NoSQL, MongoDB, Redis, Kubernetes, Git, Prompt Engineering, LLM, Langchain, LlamaIndex, OpenAI, Faiss, ChromaDB, Weaviate.*

Projects

- MyTripMate: AI-powered Trip Planner** | *Python, Google ADK, Vertex AI, Google Cloud Platform* [↗](#)
- **Built a multi-agent AI travel planner** using Google ADK, Vertex AI, and Google Cloud Platform with autonomous workflows for itinerary generation, accommodation planning, and booking automation.
 - Designed automated travel, accommodation planning based on budget, days, weather, and safety insights and booking workflows with one-click execution.
 - Submitted the prototype at Google AI Hackathon 2025, demonstrating expertise in AI-driven travel automation.
- Resume Screening & ATS Scoring Platform** | *Python, Langchain, FAISS, OpenAI, Gemini, Groq, HuggingFace, ReAct* [↗](#)
- Architected a modular multi - LLM Resume Screener achieving <3s parsing-to-scoring latency per candidate profile.
 - Engineered a file-based FAISS RAG pipeline indexing 50+ resumes/batch with sub-50ms semantic search response time.
 - Built a conversational ReAct Agent that automates candidate audits and matches skills against 3k+ character JDs.
- ByteOn AI** | *React.js, Node.js, Nodemailer, Google Sheet, Calendly, Tailwind CSS* [↗](#)
- Developed a fully dynamic, end-to-end web application featuring Framer Motion animations to deliver a smooth, interactive, and visually engaging user experience.
 - Integrated Calendly API to enable real-time meeting scheduling, for seamless client meeting scheduling.
 - Automated lead management by integrating Google Sheets and enable direct email communication.
- Photo-Realistic Single Image Super Resolution Using GAN** | *Python, Numpy, PyTorch and celebA Dataset*
- Implemented SRGAN architecture aimed at enhancing the finer details of degraded images.
 - Trained a Generative Adversarial Network (GAN) to generate up to 4-times upscaled high-resolution image.
 - Optimized model performance by fine-tuning hyperparameters and utilizing perceptual loss to enhance image quality.

Education

Indian Institute of Technology Jammu, Jammu <i>Master of Technology in Computer Science and Engineering</i>	Sep 2021 - Jun 2023 <i>CGPA - 8.25/10</i>
Nalanda College of Engineering, Chandi, Bihar <i>Bachelor of Technology in Computer Science and Engineering</i>	Aug 2017 - Sep 2021 <i>CGPA - 8.28/10</i>

Achievements

- Ranked in the top **4%** among **GATE 2021** Computer Science candidates, showcasing exceptional aptitude.
- Attained **top 3** position in B.Tech third year, earning prestigious Academic Excellence Award.
- Achieved **under 25th rank** in the HCL First Career Grand Quest 2021 [↗](#) , a national-level competition.
- Secured **3rd position** in TEQUIP3 hackathon organized by LNJPIT, Chapra, Bihar, demonstrating coding prowess.
- Demonstrated DSA problem-solving skills over **300 challenges** on Platforms like **GeeksForGeeks** [↗](#) , **Leetcode** [↗](#) .

Position of Responsibilities

Teaching Assistant IIT Jammu , <i>Jammu and Kashmir</i>	Sep 2021 - May 2023
• Assisted Assistant Prof. Harkeerat Kaur in Design Analysis Algorithm Lab assignments and exam evaluations. • Supported Associate Prof. Bruhadeshwar Bezawada with Operating System Lab assessments, and evaluations.	